Universidade Politécnica Bragança
Escola Superior Agrária

UTFPR
UNIVERSIDADE TECNOLÓGICA FEDERAL DO PARANÁ
CAMPUS DOIS VIZINHOS

Anaerobic Co-digestion of Food Waste and Glycerol in Single-Stage System

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Introduction

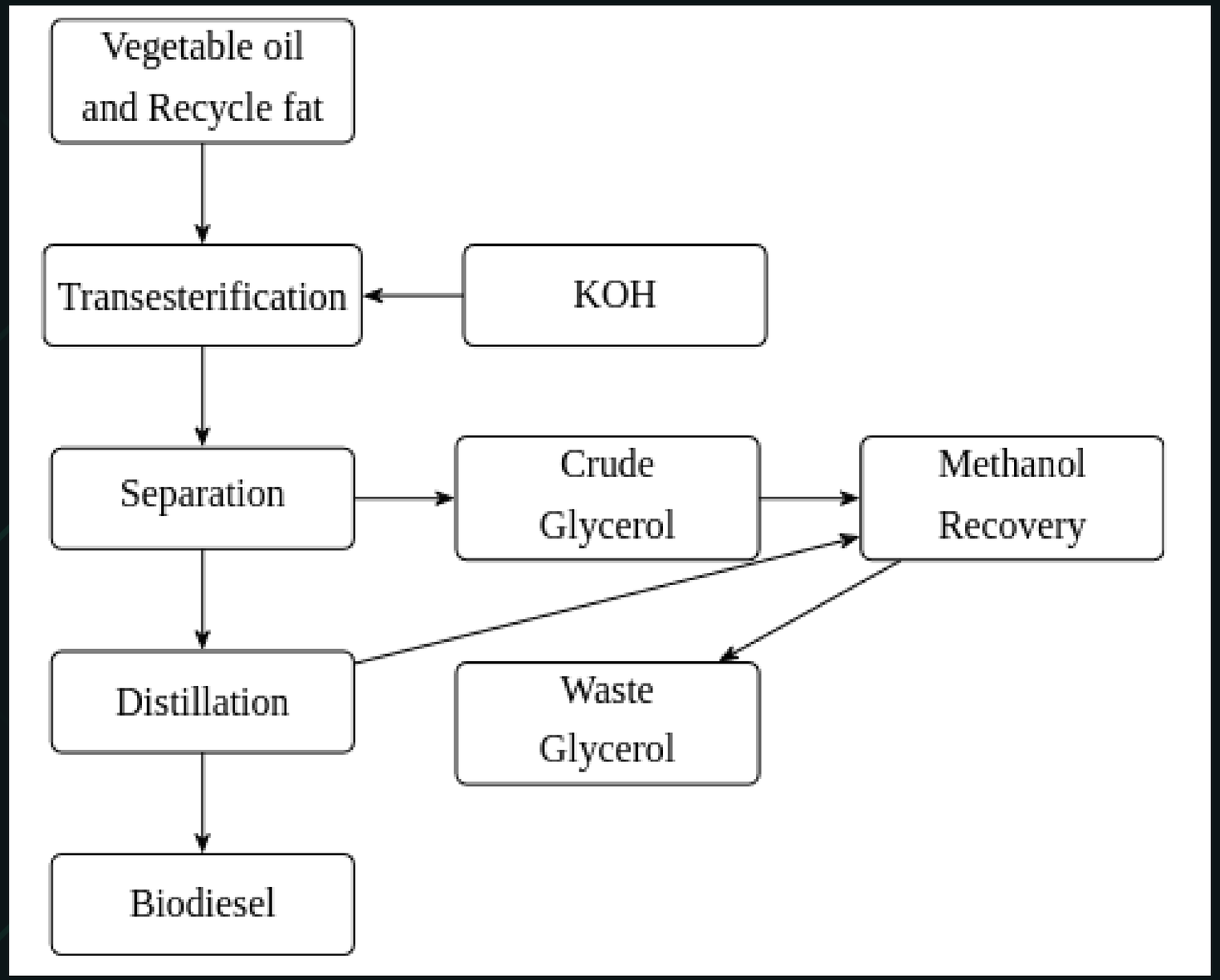
- Europe is estimated to produce 88 Mtonnes per year
- EU-27+1 66.3 Mtonnes per year of food waste.
- 186 Mtonnes CO2 - 15 to 16% of total climate impact

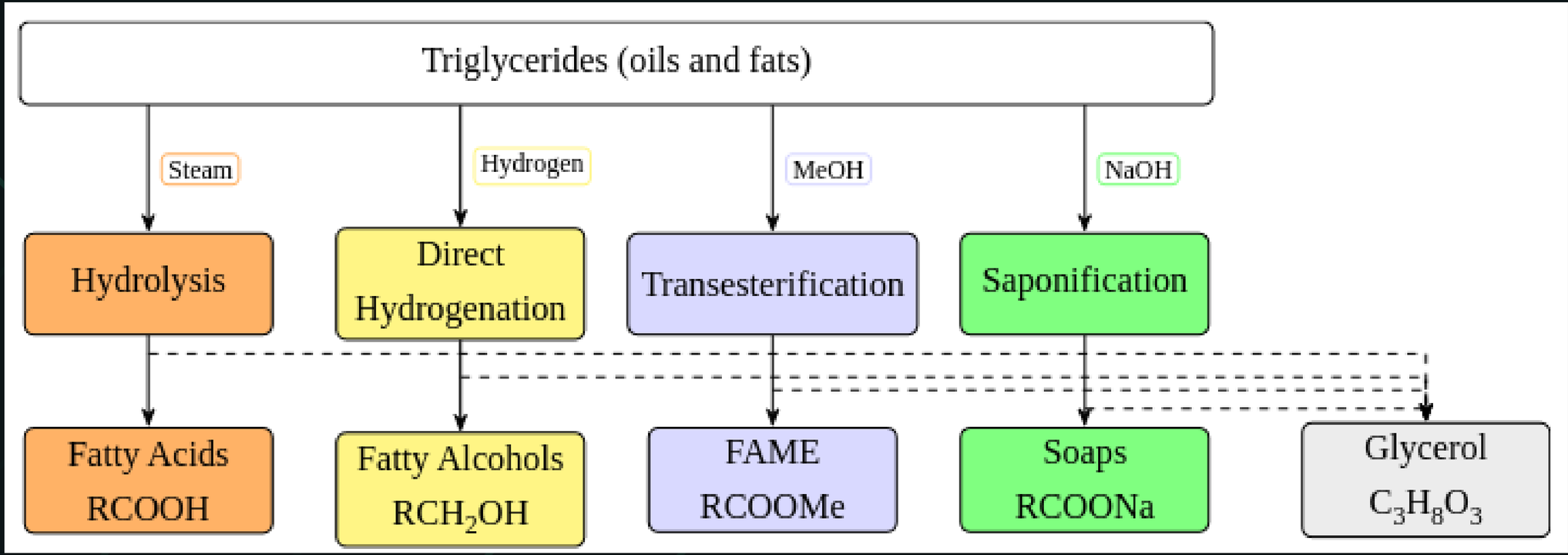
Introduction

Approximately 10 Kg of crude glycerol is produced for every 100 Kg of biodiesel.

With the changes to renewable energies approximately 6.33 Mtonnes of crude glycerol is to be produced, which 680 Ktonnes are too contaminated to be used.

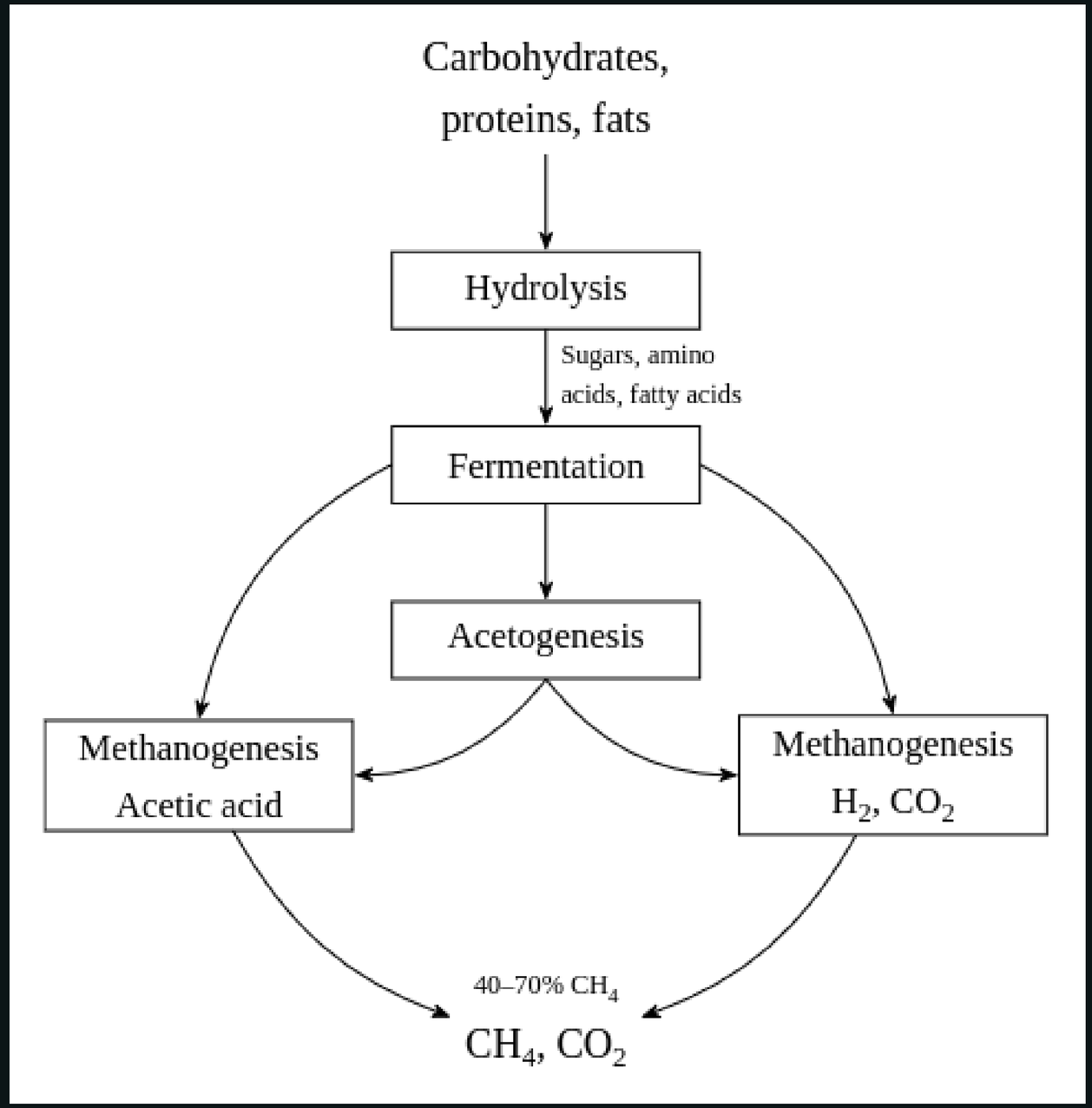
Glycerol is produced at two points of the biodiesel production





The glycerol is contaminated with fatty acids, alcohols and soaps.

Anaerobic digestion produces primarily methane in a oxygen free condition.
An excessive amount of glycerol could lead to problems maintaining the right conditions for the anaerobic digestion.



Objectives

- Characterize the substrates and evaluate the physicochemical parameters relevant to anaerobic digestion.
- Operate the single-stage anaerobic co-digestion reactor and assess process stability.
- Efficiency evaluation of the process for the utilization of different substrates and organic loading rates.

Materials and Methods

- Preparation of Substrate and Collection of Inoculum
- Physical and Chemical Characterization
- Nutrients

Sample		$M_{crucible}$ (g)	M_{wet} (g)	M_{sample} (g)	V_{sample} (L)	M_{105} (g)	TS (g L^{-1})	Average TS (g L^{-1})	M_{550} (g)	VS (g L^{-1})	Average VS (g L^{-1})
Food Waste	1.000	7.873e1	1.060e2	2.726e1	2.550e−2	9.100e1	4.810e2	5.364e2	7.920e1	4.625e2	5.208e2
	2.000	8.432e1	1.133e2	2.898e1	2.710e−2	9.887e1	5.364e2		8.474e1	5.208e2	
	3.000	7.759e1	1.088e2	3.125e1	2.920e−2	9.344e1	5.420e2		7.818e1	5.216e2	
Crude Glycerol	1.000	8.422e1	8.864e1	4.424	5.000e−3	8.850e1	8.634e2	8.677e2	8.421e1	8.644e2	8.687e2
	2.000	7.348e1	7.799e1	4.509	5.000e−3	7.786e1	8.677e2		7.347e1	8.687e2	
	3.000	8.620e1	9.061e1	4.410	4.900e−3	9.050e1	8.713e2		8.620e1	8.724e2	
Inoculum	1.000	8.991e1	9.745e1	7.537	7.600e−3	8.997e1	6.130e−1	5.244e−1	8.992e1	4.620e−1	3.643e−1
	2.000	8.017e1	8.873e1	8.564	8.600e−3	8.022e1	5.244e−1		8.018e1	3.643e−1	
	3.000	7.843e1	8.679e1	8.356	8.400e−3	7.848e1	4.663e−1		7.845e1	3.105e−1	

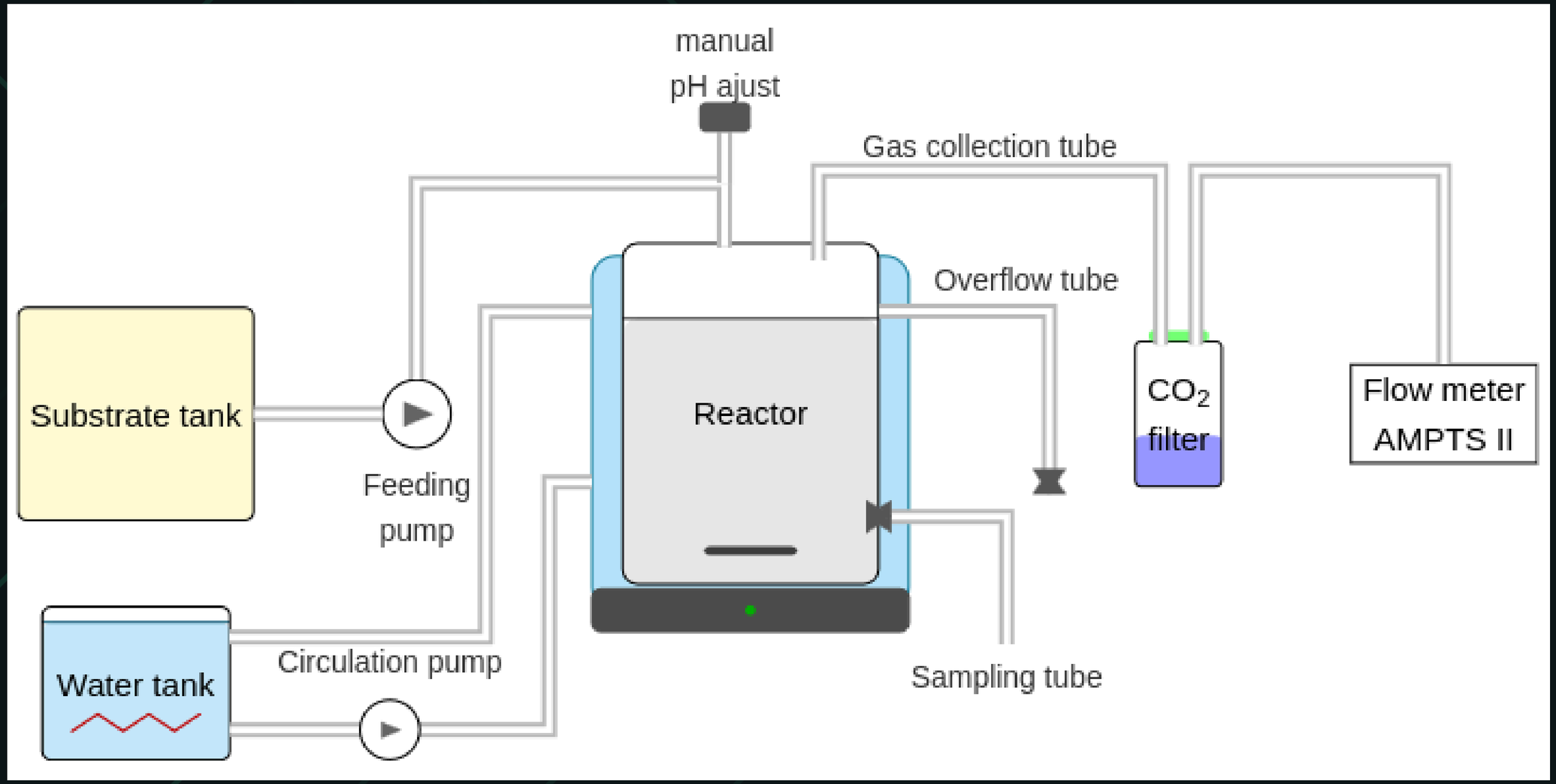
Sample		$M_{cylinder}$ (g)	M_{total} (g)	V_{sample} (L)	Density (g L^{-1})	Average Density (g L^{-1})
Food Waste	1.000	1.360e2	2.420e2	9.920e−2	1.069e3	1.068e3
	2.000	1.160e2	2.230e2	1.002e−1	1.068e3	
Crude Glycerol	1.000	1.142e2	2.035e2	9.990e−2	8.932e2	8.931e2
	2.000	1.157e2	2.044e2	9.930e−2	8.929e2	
Inoculum	1.000	1.325e2	2.314e2	9.920e−2	9.978e2	9.950e2
	2.000	1.157e2	2.150e2	1.000e−1	9.922e2	

Sample		TOC (mg L^{-1})	TC (mg L^{-1})	Average TC (mg L^{-1})	TN (mg L^{-1})	Average TN (mg L^{-1})
Food	1.000	5.039	2.018e3	1.779e3	2.756e2	2.818e2
Waste	2.000	3.927	1.540e3		2.879e2	
Crude	1.000	8.808	7.050e3	5.523e3	4.700e2	4.886e2
Glycerol	2.000	4.992	3.997e3		5.072e2	

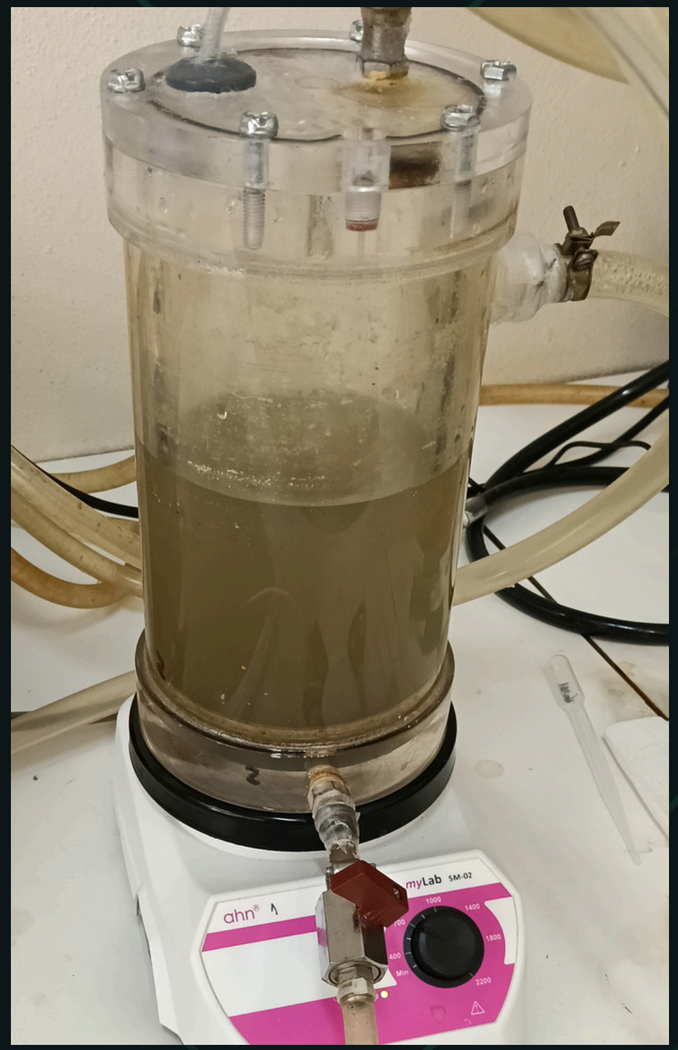
Nutrients	Concentration (mg L^{-1})
$MgSO_4 \cdot 7\,H_2O$	5.400e1
$FeSO_4 \cdot 7\,H_2O$	1.494e3
$CoCl_2 \cdot 6\,H_2O$	1.200
$ZnCl_2$	3.000e1
$NiCl_2 \cdot 6\,H_2O$	3.000e−2
EDTA	6.000e−1
$CuSO_4 \cdot 5\,H_2O$	2.637e1




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manual
pH ajust
Gas collection tube
Overflow tube
Substrate tank
Feeding
pump
Reactor
CO_2
filter
Flow meter
AMPTS II
Water tank
Circulation pump
Sampling tube

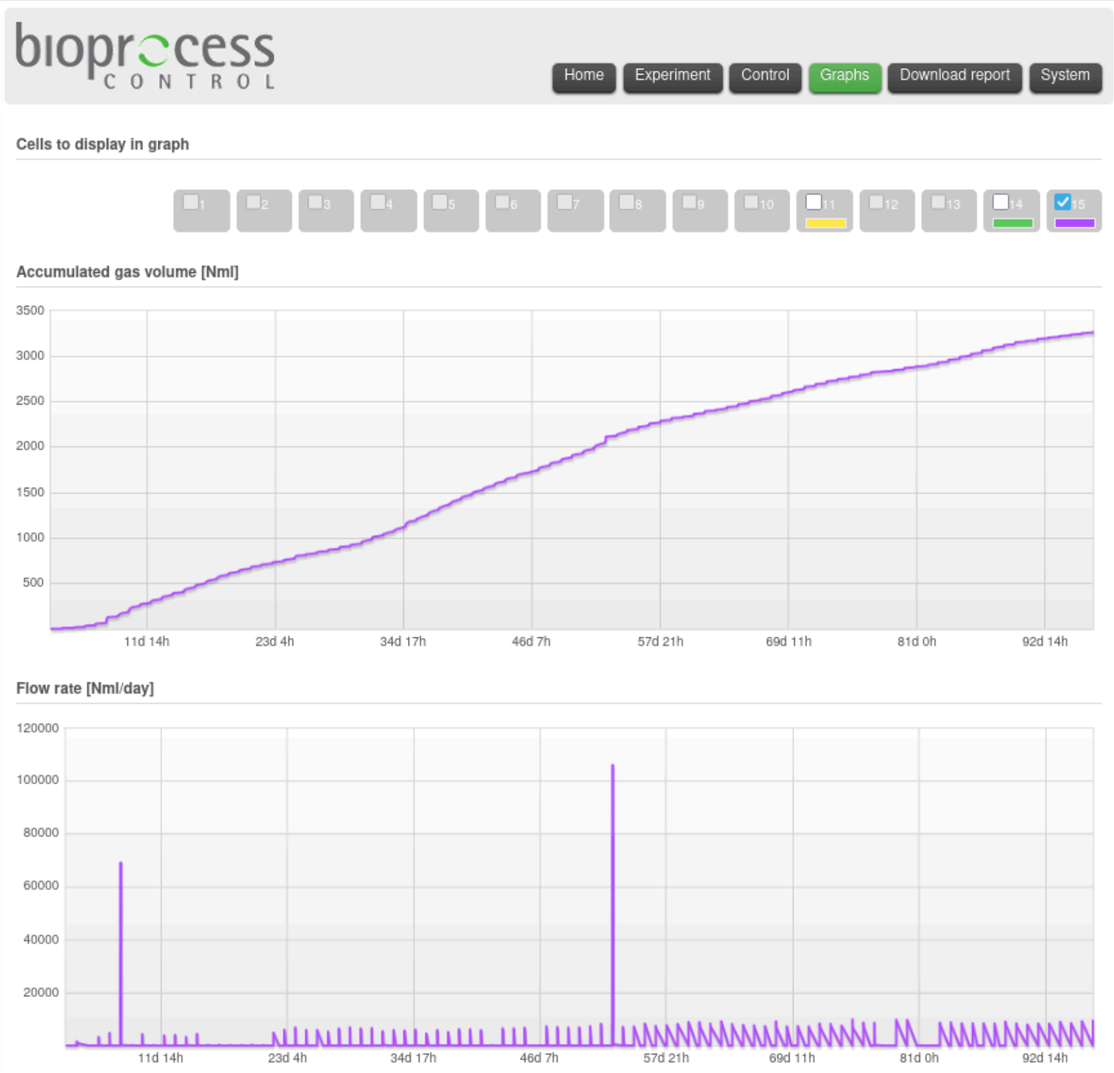
Results and Discussion

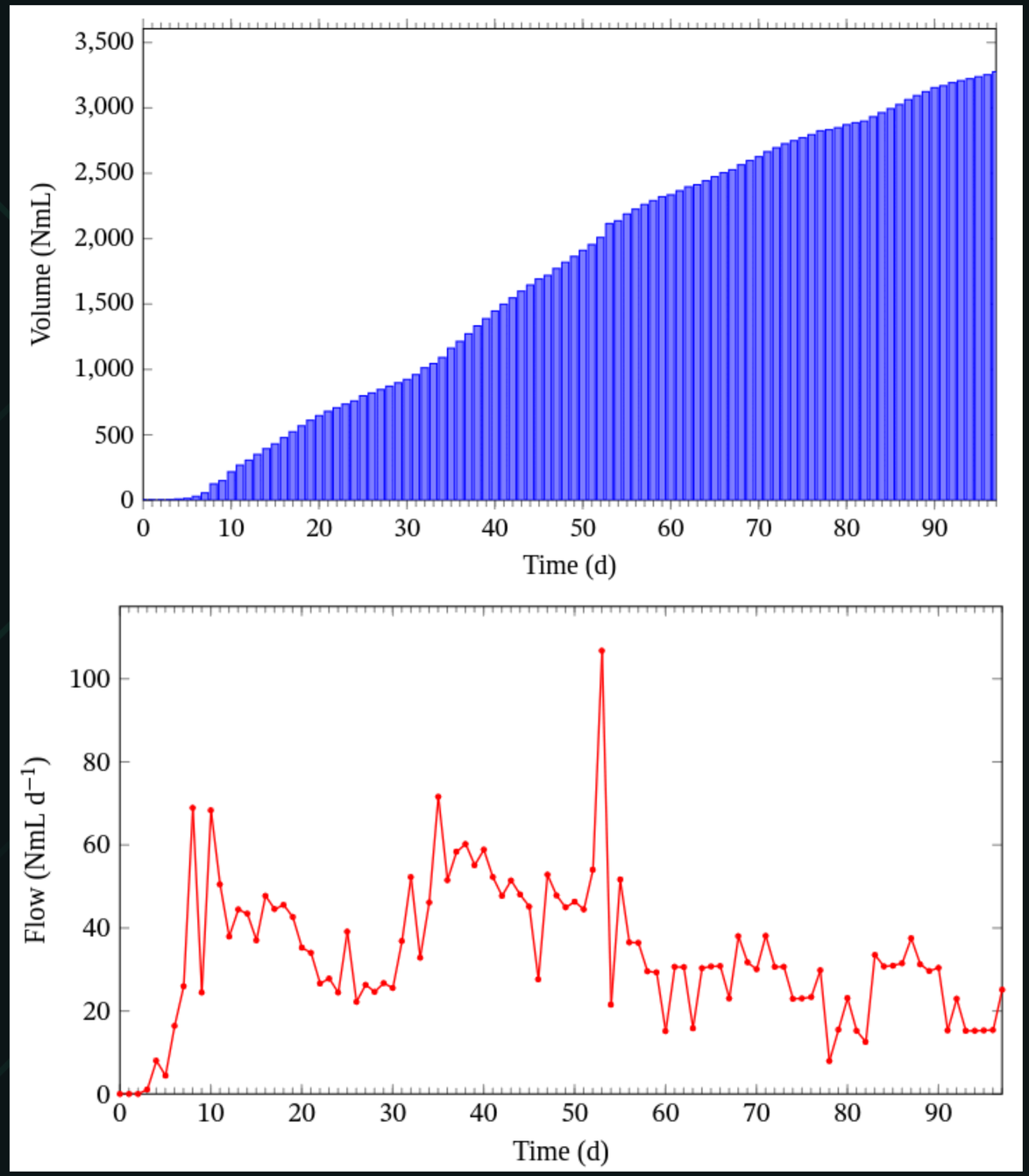






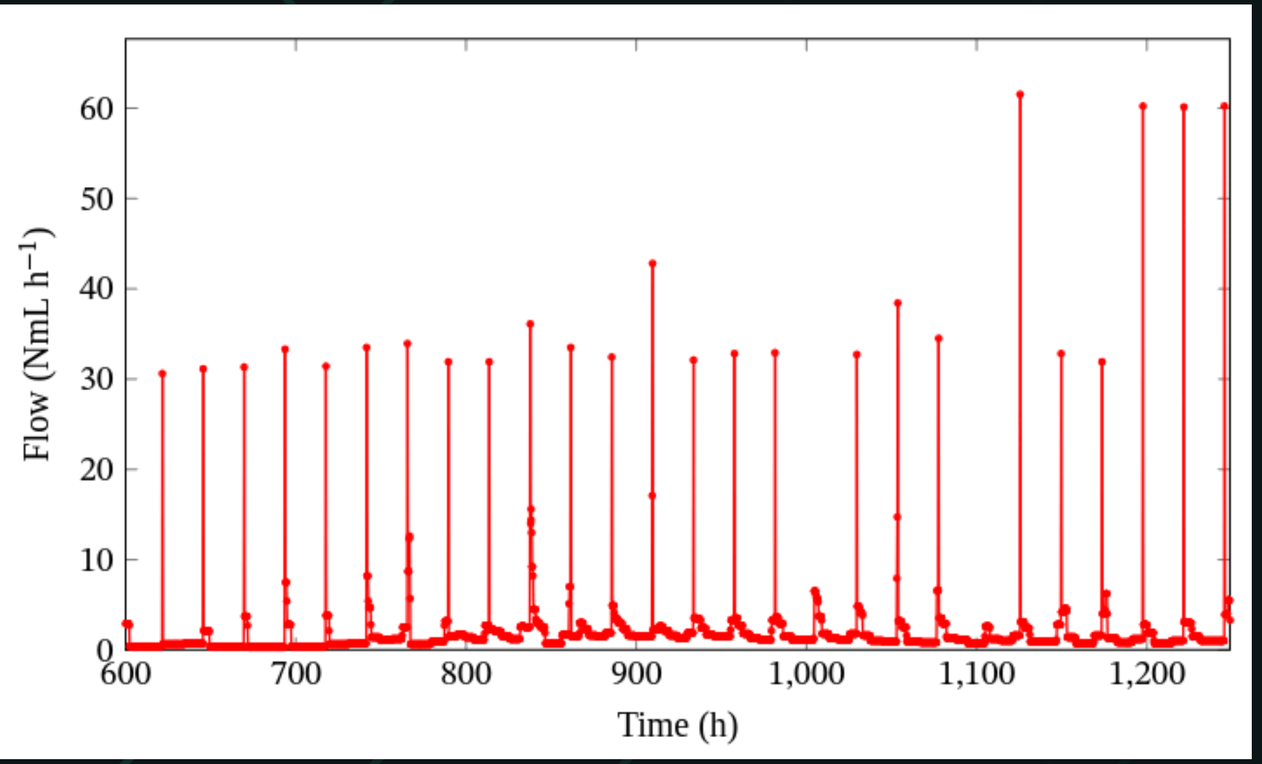




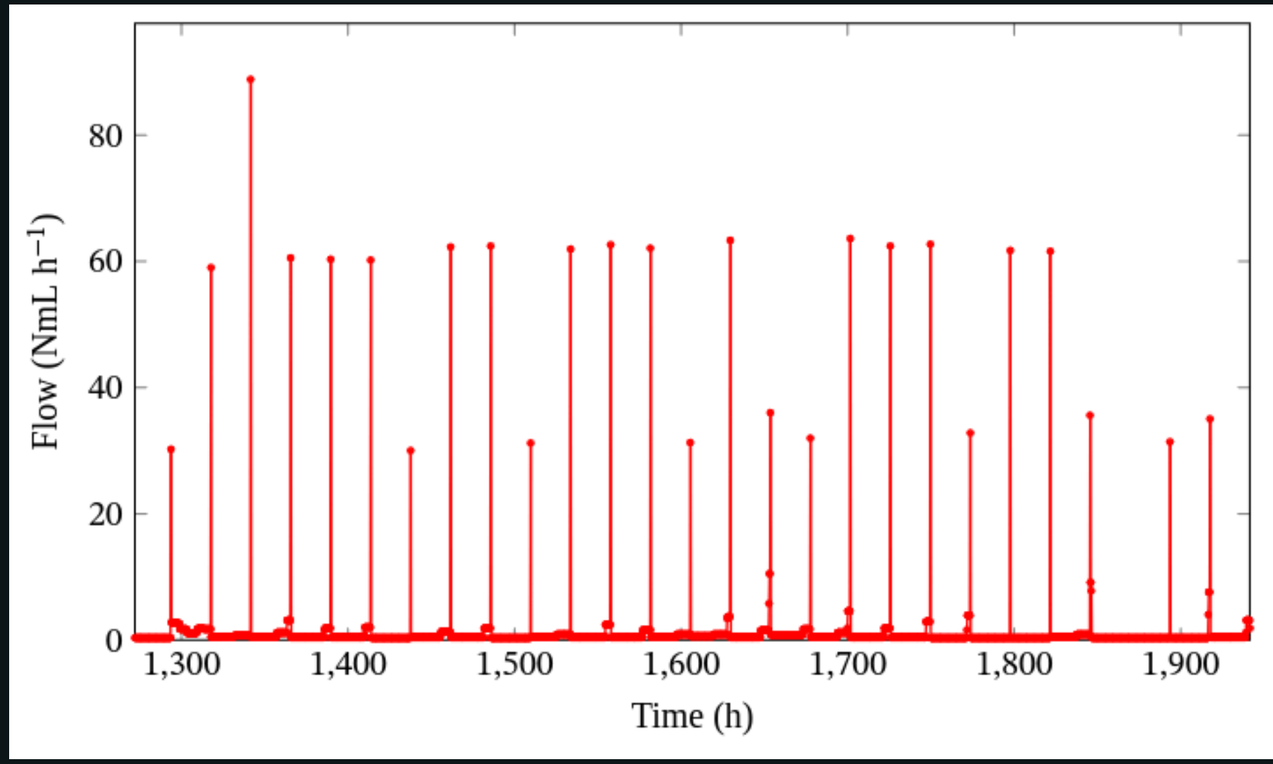


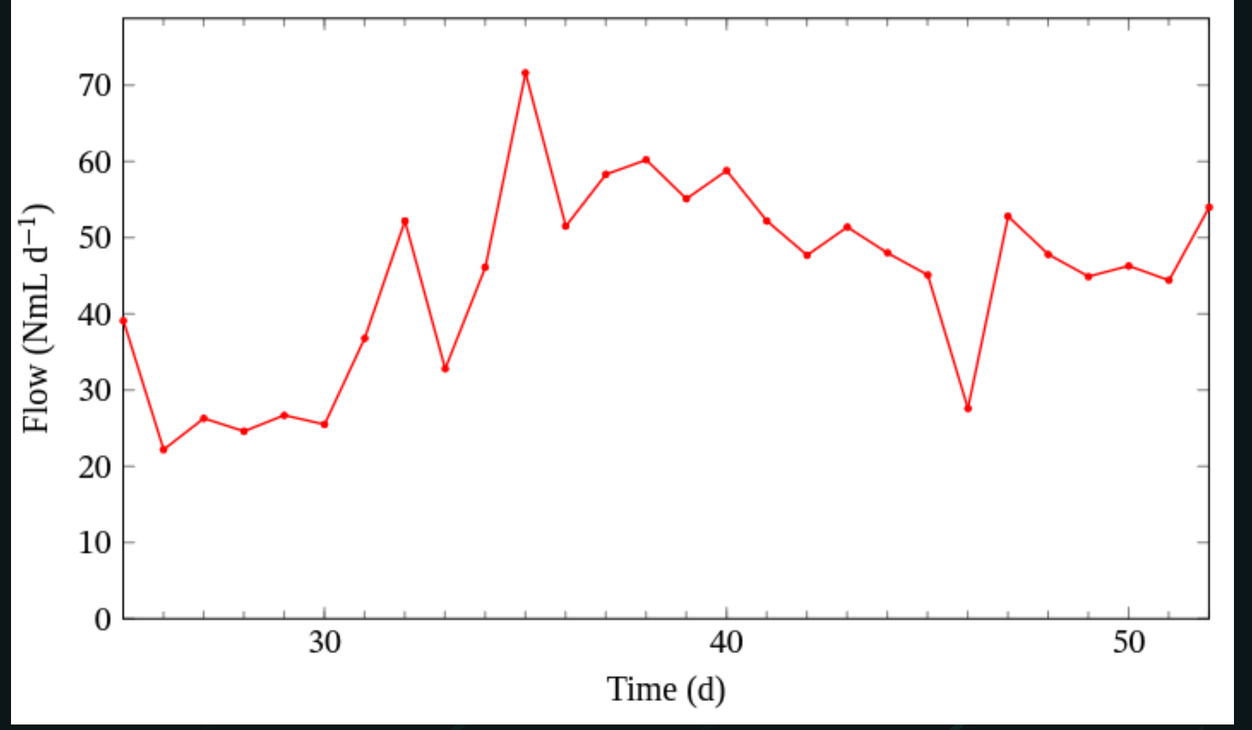
Total accumulation and production of Methane

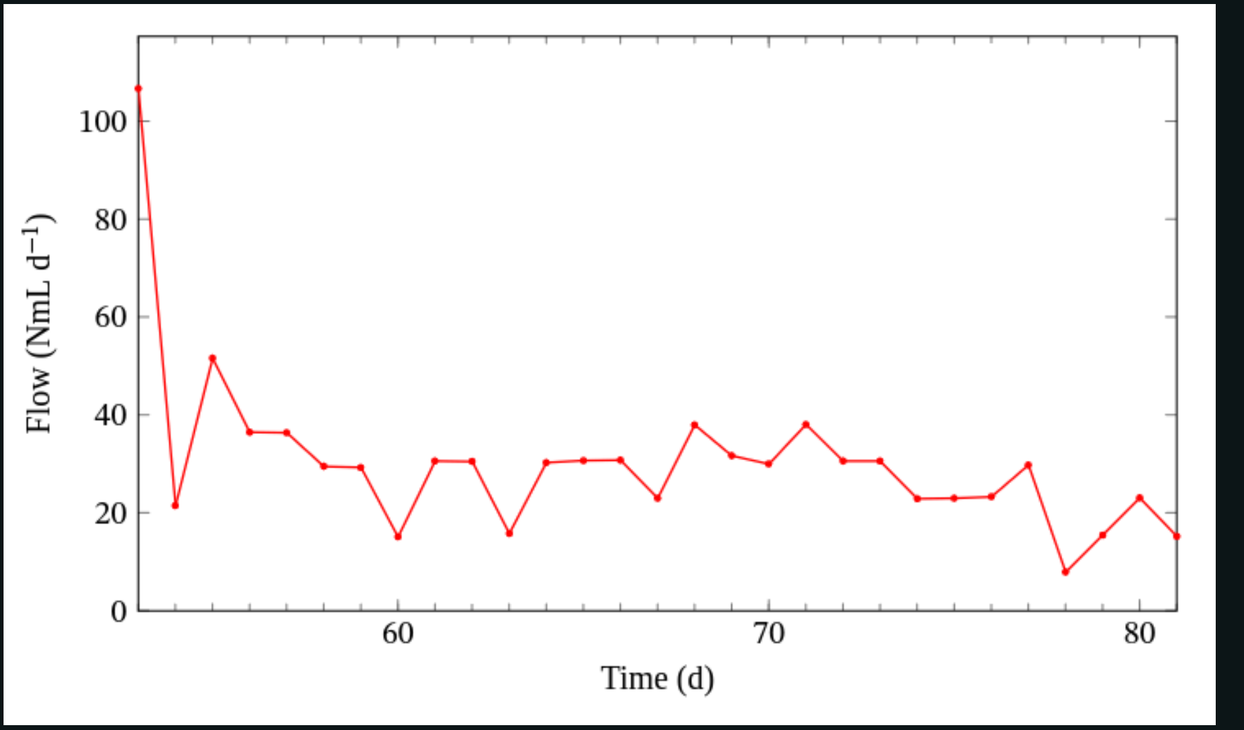
Flow of Methane during co-digestion

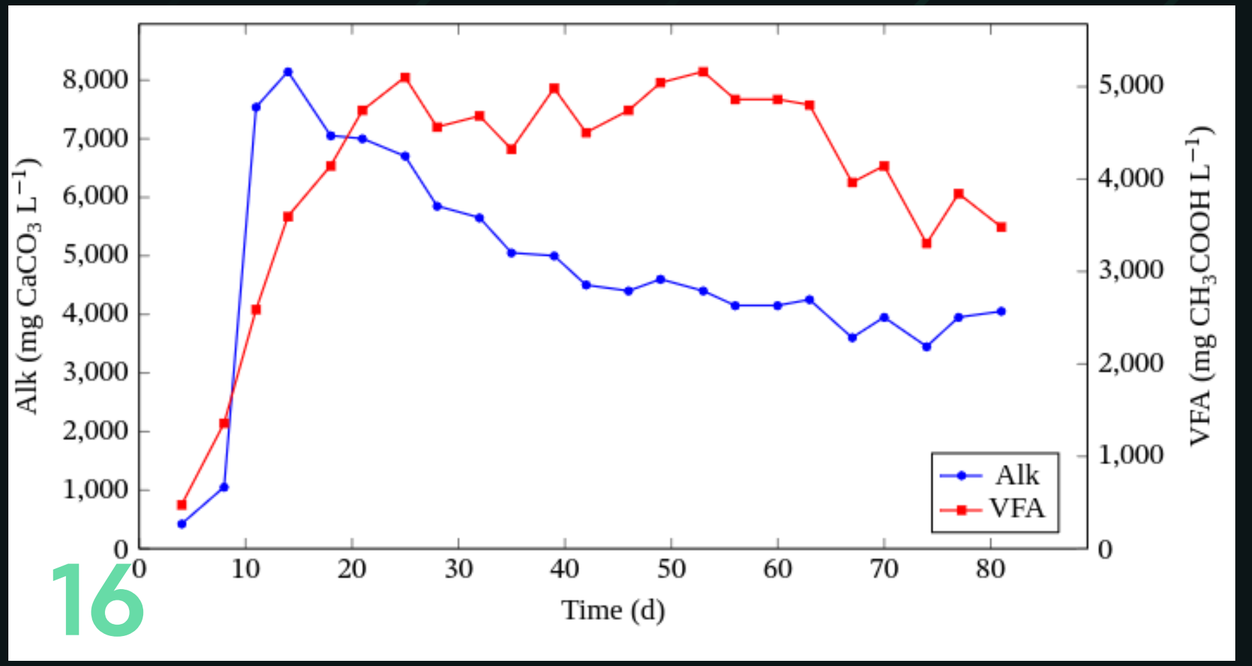


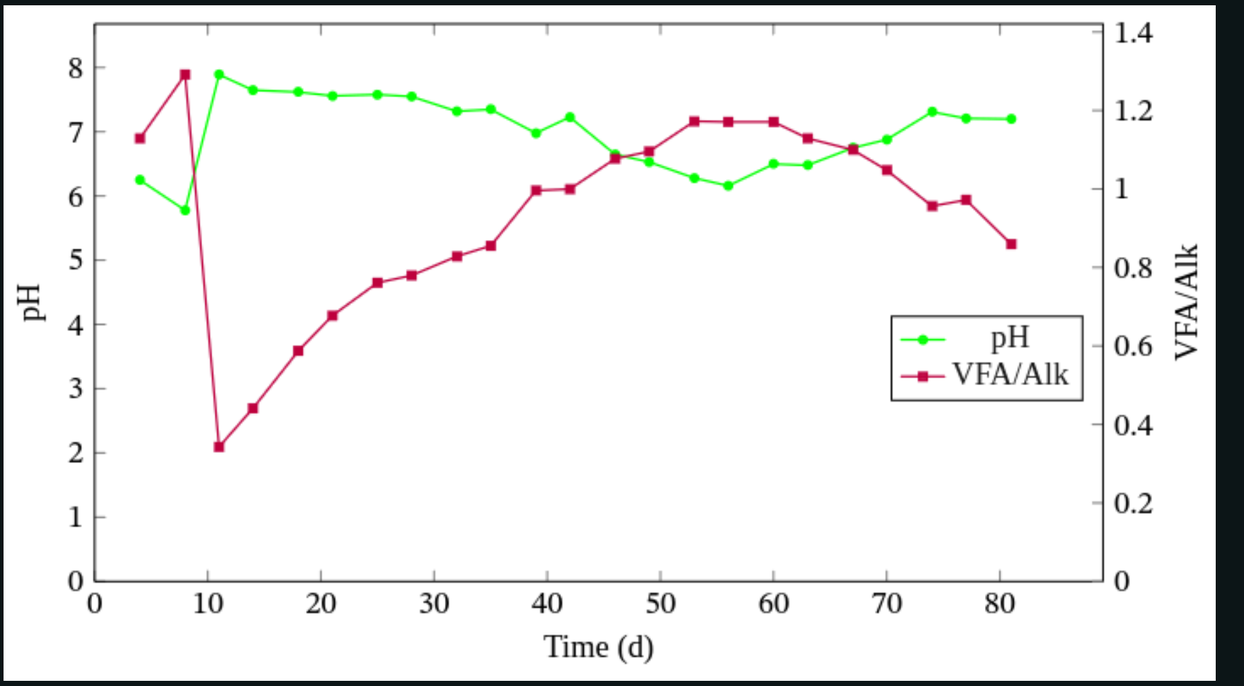
Flow of Methane during Crude Glycerol phase

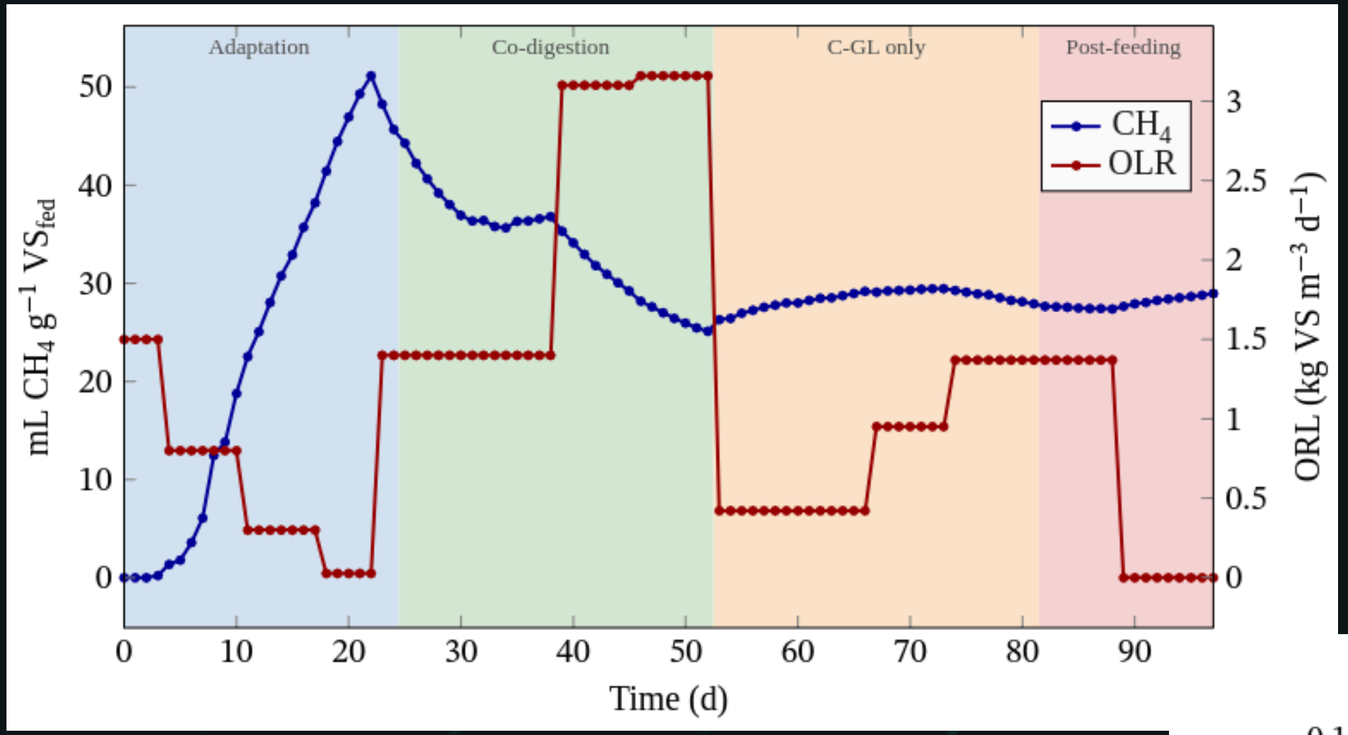



Flow (NmL d^{-1})
Time (d)


Flow (NmL d^{-1})
Time (d)

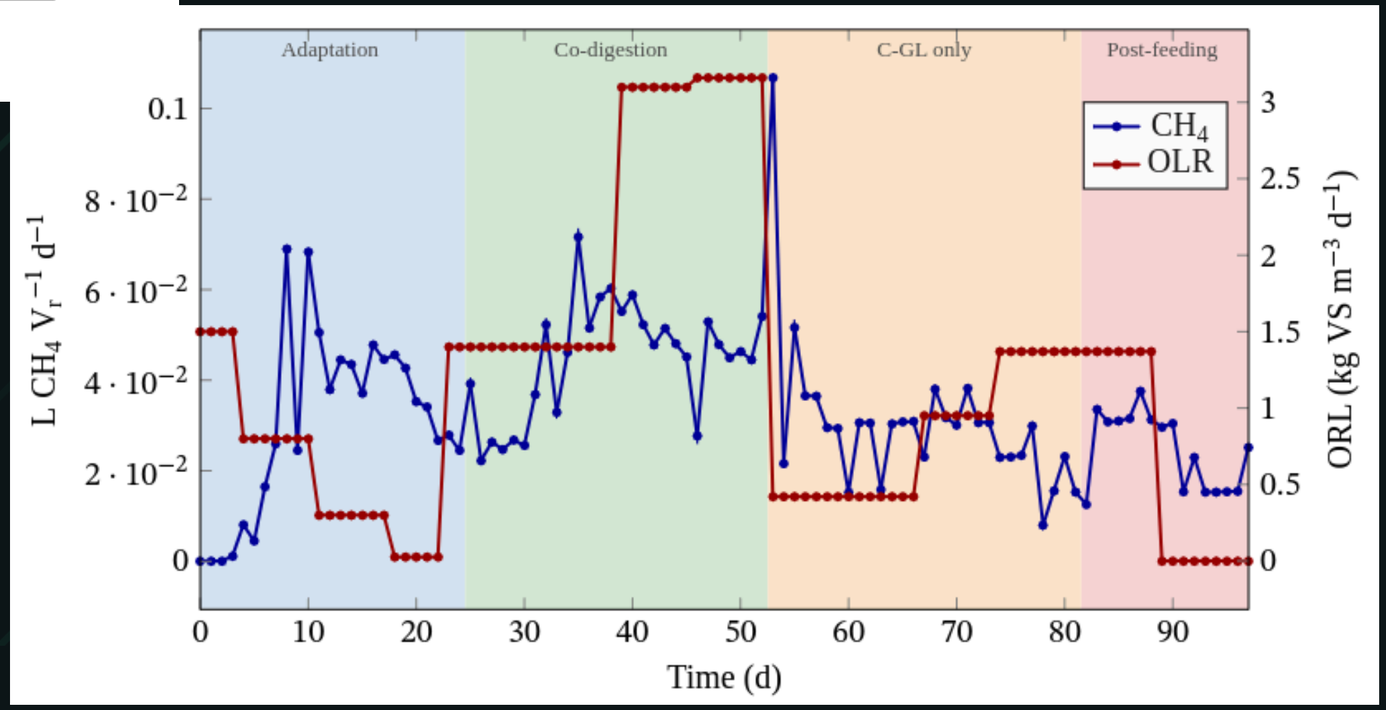

Alk (mg $CaCO_3$ L^{-1})
VFA (mg CH_3COOH L^{-1})
Time (d)
Alk
VFA


pH
VFA/Alk
Time (d)
pH
VFA/Alk



Methane production related to volatile solids feed, and organic loading rate

Methane production related to the reactor volume per day, and organic loading rate



Conclusion

- Higher stability and pH decreased faster with co-digestion process
- Crude Glycerol phase shouldn’t be considered worse, by default
- It was needed to use pure glycerol to balance the C/N, for the impurities on the crude glycerol
- VFA/ALK of 1 and pH of 6.5 showed the best performance
- The average production was 20 NmL/g VS for the co-digestion (food waste and glycerol) and 37 NmL/g VS for glycerol only

Thank you